Aligning Metabolomic Networks via Graph Learning for Common Subnetwork Discovery

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Motivation

Mass spectrometry is an analytical technique that is used to identify and quantify chemical compounds in a sample.

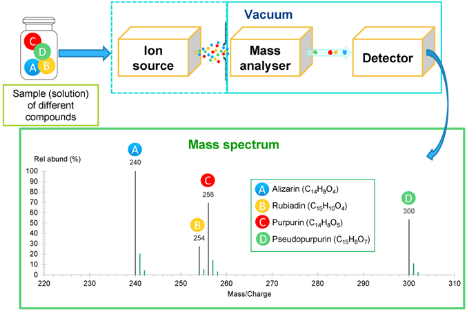


Figure 1: Overall scheme of a mass spectrometer and formation of a mass spectrum.
Source: `https://sisu.ut.ee/heritage-analysis/52-mass-spectrometry`

Problem formulation

The mass spectrometry analysis includes two approach:

- **Targeted approach:** the molecule of interest or m/z values (for MS and MS/MS) are predefined for quantitation of analytes present in a complex samples matrix.
- **Untargeted approach:** is the analysis to identify the unknowns in a sample, using parent ion and product or fragment ion analysis (MS/MS).

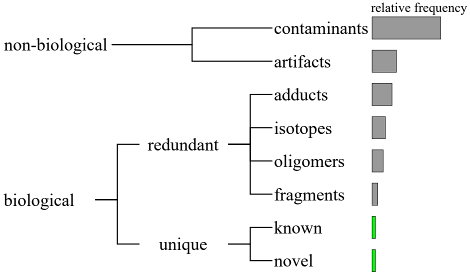


Figure 2: Composition of an untargeted metabolomic data.

Source: Adapted from [?]

Untargeted mass spectrometry workflow

A data matrix (sample vs. metabolites) acquired from a mass spectrometry-based metabolomics study includes not only biological information but also technical noise.

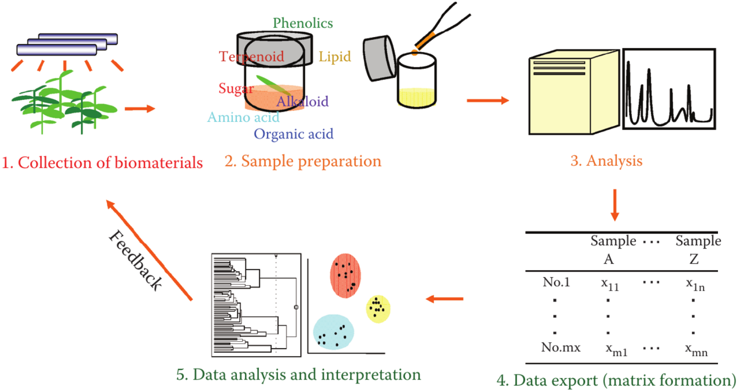


Figure 3: Metabolomics workflow.

Source: [?]

Our objectives

Aligning metabolomic networks for filter MS data.

Mass spectrometry (MS)

Mass spectrometry (MS) is a microanalytical technique used to detect and determine the amount of an analyte, it is also used to determine the elemental composition and some aspects of the molecular structure of an analyte.

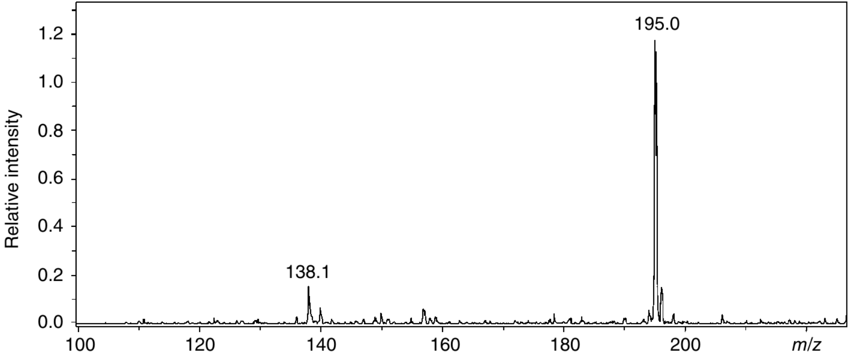


Figure 4: Mass spectrum of caffeine obtained by electrospray ionization mass spectrometry (ESI-MS) technique.

MS dataset format

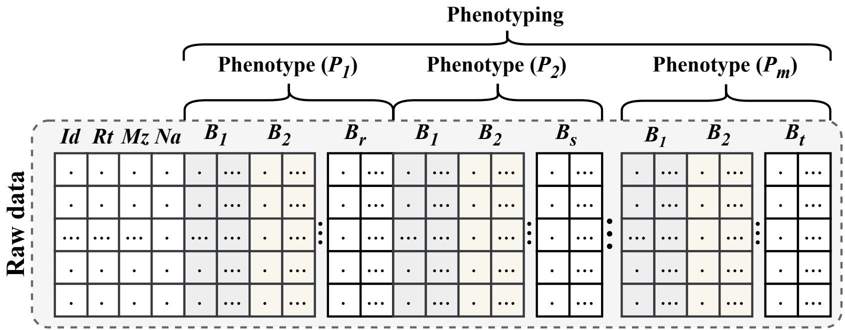
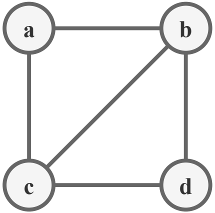


Figure 5: Aligned mass spectrometry data format.

Graph

A graph (network) is defined as $G = (V, E)$, where $V = \{v_1, v_2, \ldots, v_n\}$ is the nodes (or *vertices*, or *points*) set and $E = \{e_1, e_2, \ldots, e_m\}, e_i \in V \times V$ a set of edges (or *lines*, or *arcs*) formed by unordered pairs of nodes.



(a) Input graph.

	a	**b**	**c**	**d**
a	0	1	1	0
b	1	0	1	1
c	1	1	0	1
d	0	1	1	0

(b) Adjacency matrix.

a	b	c	
b	a	c	d
c	a	b	d
d	b	c	

(c) Adjacency list.

a	b
a	c
b	c
b	d
c	d

(d) Edge list.

Figure 6: Graph representations.

Graph Neural Network (GNN)

It is a function $f : (A, X) \to Z$ that maps each node $v \in V$ of a graph $G = (V, E)$ to a low-dimensional embedding $z_v \in \mathbb{R}^d$ $(d \ll |V|)$.

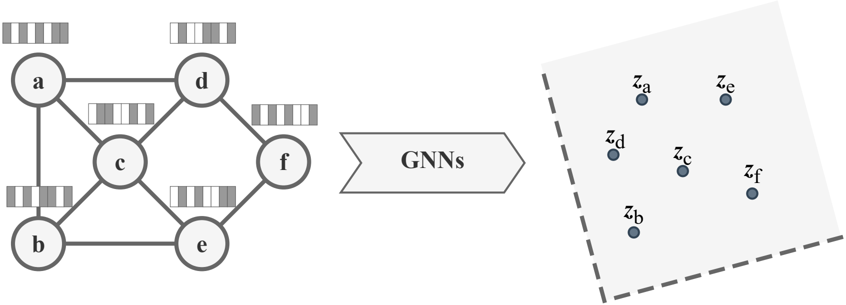


Figure 7: GNNs scheme.

Graph aligment problem

Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two graphs. The goal is to align their nodes through a mapping function $\pi : \mathbf{V_1} \rightarrow \mathbf{V_2}$, where each node $v \in V_1$ is mapped to its corresponding node $\pi(v) \in V_2$.

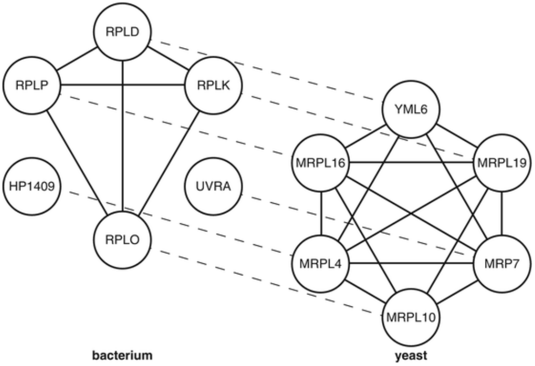


Figure 8: Node alignment.

Source: https://link.springer.com/rwe/10.1007/978-1-4419-9863-7_992

Base GNN model

T-GAE: Transferable Graph Autoencoder for Network Alignment.
(He, J., Kanatsoulis, C., & Ribeiro, A., 2024)

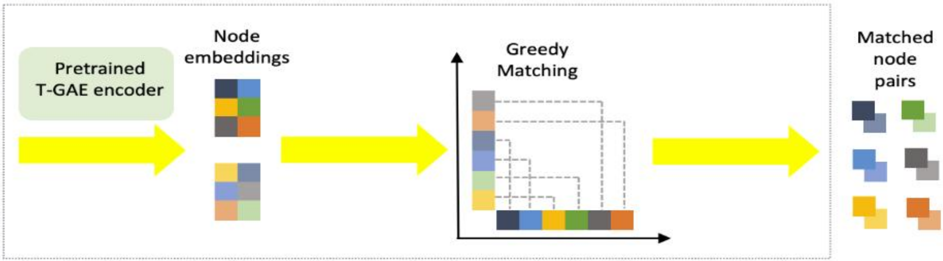


Figure 9: T-GAE encoder.

Method

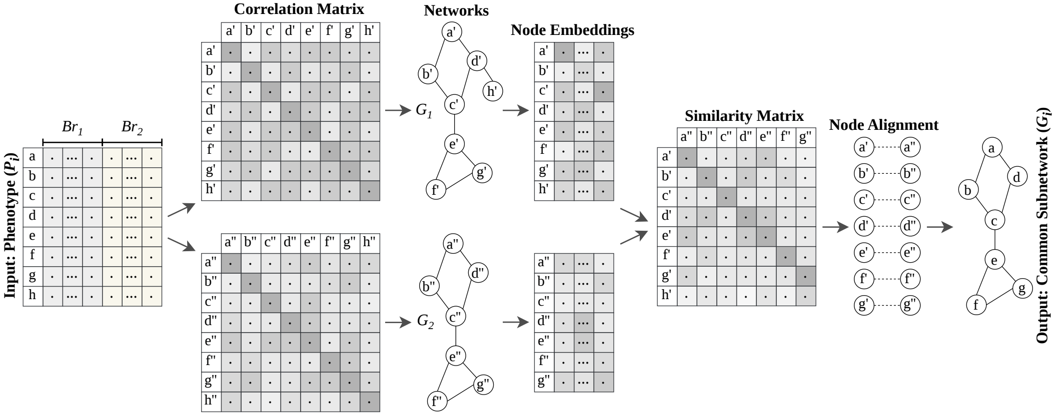


Figure 10: Our workflow.

Method overview

1. **Correlation estimation:** compute a partial correlation matrix.
2. **Network construction:** build a metabolic network using a correlation threshold ($|\text{coef}| \geq 0.5$) and incorporate node attributes (Rt, Mz, and intensities).
3. **Graph representation learning:** learn node embeddings with the `T-GAE` architecture based on `GIN` and `GINE` layers.
4. **Network alignment:** identify metabolite correspondences using a `KNN`-based similarity matrix.

Setup

The experiments were conducted on a computer with 256 GB of RAM and 48 GB of VRAM. The tools used were `Python` and the `PyTorch Geometric` library.

Dataset

Table 1: Datasets.

Dataset	Type network	#Nodes	#Edges	#Features	#Anchors/Pairs
Douban Online	Social	3 906	1 632	538	1 118
Douban Offline	Social	1 118	3 022	538	1 118
ACM	Citation	9 916	44 808	17	6 325
DBLP	Citation	9 872	39 561	17	6 325

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4. Experiments
5. Results
6. Conclusions

Build metabolomic networks

Table 2: Dataset information.

Group	Subgroup	$\lvert V\rvert$	$\lvert E\rvert$	Density	Connected
secoAmazonas	1	2071	1 113 277	0.52	True
secoAmazonas	2	2071	1 085 089	0.51	True
secoSanMartin	1	2069	1 082 335	0.51	True
secoSanMartin	2	2071	1 103 292	0.51	True
secoCusco	1	2072	1 091 447	0.51	True
secoCusco	2	2072	1 086 182	0.51	True
frescoCusco	1	2071	1 114 859	0.52	True
frescoCusco	2	2072	1 089 225	0.51	True

Node embedding (T-GAE)

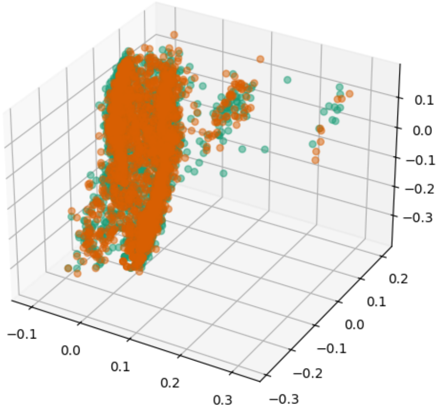

Figure 11: Node embedding (*secoSanMartin*).

Network alignment

Table 3: Node aligment (*secoSanMartin*).

ID 1	ID 2
13	13
19	252
27	378
31	1148
38	38
43	273
49	72
50	50
51	366
53	241
73	270
...	...
1825	1726

Table 4: Node filtered **133/2072** (*secoSanMartin*).

ID	Rt	Mz	Metabolite name
13	6.10	281.05	Unknown
38	6.16	182.10	Unknown
50	6.20	354.10	Unknown
128	6.47	119.20	Unknown
184	6.76	341.14	Unknown
261	7.23	371.06	Unknown
272	7.29	327.01	Unknown
314	7.61	57.02	DECANE; EI-B; MS
336	7.77	53.96	Unknown
339	7.81	328.01	Unknown
400	8.31	109.08	CITRONELLYL ACETATE...
402	8.31	109.13	BETA-HOMOCYCLOCITRAL...
407	8.37	401.11	Unknown

Thanks!

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`https://github.com/win7`

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